

HW 2

1. a) $\frac{\sin t}{t^2+2} \delta(t) = \frac{\sin 0}{0^2+2} \delta(t) = 0 \cdot \delta(t)$

b) $\frac{\cos(t)}{t^2+2} \delta(t-3) = \frac{\cos 3}{9+2} \delta(t-3) = \frac{\cos 3}{11} \delta(t-3)$

c) $\frac{2+jt}{3+(jt)^2} \cdot \delta(t) = \frac{2}{3} \delta(t)$

d) $x(t)$

e) $x(t)$

f) $\int_{-\infty}^{\infty} \delta(a\tau) d\tau = \frac{1}{a} \int_{-\infty}^{\infty} \delta(a\tau) da\tau = \frac{1}{a}$

2. a) dynamic non-causal non-invertible

stable

b) static causal non-invertible

stable

c) dynamic non-causal invertible

unstable

d) dynamic causal non-invertible

stable

e) dynamic causal non-invertible

stable

f) dynamic non-causal invertible

stable

g) dynamic causal

non-invertible unstable